

Claude Science & the Lab-in-the-Loop

AI-for-Science Stack

A technology-first teardown of the AI-for-science agent stack — and the lab-in-the-loop technique that transfers to any domain.

As promised on Day 100, we pick up the technology pipeline with the launch that jumped the queue: Anthropic shipped Claude Science on June 30, and this week it wired in NVIDIA's BioNeMo models. Today is a hands-on teardown of the AI-for-science stack -- what an autonomous research agent is made of, how the 'lab-in-the-loop' technique actually works under the hood, and why reproducibility and dual-use are the two boundaries that decide what ships. The transferable idea: the autonomy is in the LOOP, not the robot.

60+ scientific skills & connectors in Claude Science	2 yrs -> weeks Allen Institute review-writing time cut	1,500 papers read in a single Kosmos run	Level 2-3 where self-driving labs actually sit (of 5)
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1. The AI-for-science stack -- six layers under the hood

Strip away the branding and every credible AI-for-science system in 2026 is the same six-layer stack, and it maps almost one-to-one onto the agentic primitives this series has been teaching. (1) A generalist COORDINATING agent takes the plain-language question -- the orchestrator pattern from Day 24. (2) It spawns SPECIALIST sub-agents, each scoped to a task (literature, data analysis, structure prediction), and can invoke domain agents a user has authored. (3) Those agents reach SKILLS and CONNECTORS into 60+ scientific databases -- UniProt, PDB, Ensembl, Reactome, ClinVar, ChEMBL, GEO -- each with its own schema, exposed as callable tools (the SKILL.md / MCP layer from Days 55-60). (4) They call BIOMOLECULAR MODELS as tools rather than being those models -- Evo 2 for genomics, Boltz-2 and OpenFold3 for structure. (5) A COMPUTE MANAGER drafts a plan, asks before reaching new resources, and submits jobs, scaling from one GPU to hundreds on your laptop, HPC over SSH, or Modal on demand. (6) A REVIEWER / CRITIC agent checks citations and calculations and self-corrects, and every output carries an auditable trail.

Stack layer	What it does	Series primitive
1. Coordinating agent	Takes the plain-language question, plans, delegates	Orchestrator (Day 24)
2. Specialist sub-agents	Scoped to lit search, data analysis, structure	Sub-agents / harness
3. Skills + connectors	Query 60+ DBs: UniProt, PDB, ChEMBL, GEO...	SKILL.md / MCP (Days 55-60)
4. Callable models	Evo 2, Boltz-2, OpenFold3 as tool calls	Model-as-tool (Day 34)
5. Compute manager	Plans + submits jobs, laptop -> HPC -> Modal GPUs	Scoped actuation
6. Reviewer / critic	Checks citations & math; auditable artifact trail	Evaluator (Day 23)

The layer that turns this from a demo into a workbench is layer six. It is the Three-Agent Harness from Day 23 -- Planner, Generator, Evaluator -- re-cast for science, where the Evaluator is a reviewer agent whose entire job is to catch a hallucinated citation or a number that doesn't trace back to the code that produced it. That is the difference between 'an LLM wrote a plausible methods section' and 'a result you can reproduce.'

2. Claude Science, deconstructed

Claude Science (beta, June 30, on Pro/Max/Team/Enterprise) is the reference implementation of that stack. You interact with one generalist coordinating agent that has 60+ curated skills pre-configured for genomics, single-cell, proteomics, structural biology and cheminformatics; it spins up sub-agents and can call specialist agents you build. A reviewer agent inspects outputs as the pipeline runs -- flagging incorrect citations, untraceable numbers, and figures that don't match their underlying code -- and fixes them in place. Crucially, every figure ships with the exact code and environment that made it, a plain-language description, and the full message history, so a result is reproducible months later. It runs locally on macOS/Linux or over an SSH/HPC login node, so large or sensitive datasets never leave the systems they already live on -- only the context each step needs is sent to Claude. And it manages compute for you, drafting a job, asking before it reaches a new resource, letting you review or revoke,

then scaling the run. You can even fork a session to compare two approaches without losing the original thread -- git-style branching for experiments.

The early usage is the proof, not the pitch. Manifold Bio used it to nominate targets end-to-end -- assessing surface expression, trafficking and safety for each tissue and ranking candidates against criteria learned from its own proprietary data. A neuroscientist at the Allen Institute built a ~20-skill 'computational review template' using actor-critic pairs (one agent writes, a separate reviewer checks accuracy and citation fidelity), collapsing a review that took up to two years into weeks -- he now has ~10 reviews, many over 100 pages. And a UCSF glioma-epidemiology lab ran comprehensive germline workups in roughly one-tenth the previous time and independently validated the results.

3. The lab-in-the-loop -- the technique that transfers

Here is the one idea to take to any domain: a model proposes an experiment, automated hardware or a human runs it, the result returns to the model, and the model proposes the next experiment. The autonomy is not in the robotics -- it is in the LOOP. FutureHouse's Robin is the clearest public instance: given a disease name, it cycles hypothesis -> proposed experiment -> data analysis -> refined hypothesis until it lands on a candidate, with human scientists doing the physical assays (see today's viral app). This is the same closed-loop structure as a self-healing agent (Day 14) or an eval-driven build loop (Day 45), just with a wet lab as the environment.

THE ONE RULE

The autonomy is in the LOOP, not the robot. Automate one tight loop where the experiment is cheap, fast and reversible — and keep a human on the physical step and on the 'is this real?' call.

A reality check keeps this honest. Most 'self-driving labs' today sit at Level 2-3 on a five-level autonomy scale -- closed-loop optimisation on narrow tasks (reaction optimisation, materials screening), not general-purpose science. Materials and chemistry lead; drug discovery is close behind (Recursion, Arctoris); synthetic biology is the most ambitious, with Ginkgo committing all R&D; to autonomous infrastructure by end-2026. The lesson for builders: don't try to automate the whole scientific method. Automate one tight loop where the experiment is cheap, fast and reversible, and keep a human on the physical step and on the 'is this real?' call.

4. Models as callable skills -- the BioNeMo pattern

The second technique worth stealing: your agent does not need to BE a genomics model -- it needs to know when to CALL one. NVIDIA's BioNeMo Agent Toolkit (announced late June, on GitHub) turns heavyweight biomolecular models into agent-callable skills and NIM microservices: Evo 2 (genomics), Boltz-2 and OpenFold3 (structure), packaged as containerised inference endpoints with the accelerated stack pre-tuned. A single workflow might fingerprint a compound library, cluster the hits, generate conformers for the top structures, analyse genomic context, compare perturbation responses, and only then recommend the next physical experiment -- each step a tool call to a specialist model, orchestrated by a generalist agent.

This is the Day 34 multimodal insight generalised: specialist capability arrives as a tool, not a bigger brain. Both Anthropic and OpenAI are integrating the toolkit, and adopters already include Lilly, Schrodinger, Databricks, Snowflake, Dassault Systemes and the UW Institute for Protein Design. For anyone building an agent in a technical domain, the pattern is portable: wrap your best domain models as MCP skills with crisp descriptions, and let the coordinator route to them.

5. Governance -- reproducibility is the trust boundary, dual-use is the wall

In science the audit log IS the methods section (Day 76), and Claude Science makes that concrete: code + environment + plain-language description + full message history attached to every artifact means a reviewer, a journal, or a regulator can reconstruct exactly what the agent did. That same trail is the EU AI Act evidence pack -- clinical decision support is Annex III high-risk, anything published falls under Article 50 disclosure, and Aug 2 is now T-26. The other boundary is harder: the same agent that designs a therapeutic can design a hazard. That is precisely why the Mythos/Fable capability-gating from Day 99 silently reroutes bio/chem/cyber queries to a safeguarded model, and why an AI-for-science workbench must ship reviewer agents, scoped compute (the agent asks before reaching a new resource and every decision is revocable), and auditable artifacts by construction -- not bolted on.

DUAL-USE IS THE WALL

The same agent that designs a therapeutic can design a hazard — which is why Fable/Mythos capability-gating (Day 99) reroutes bio/chem/cyber queries, and why reviewer agents, scoped revocable compute, and auditable artifacts must be built in by construction, not bolted on.

Net: treat a science agent like any high-stakes agent from this series. Scope its identity and its compute (KYA, Day 54). Keep a human on the physical experiment and the final scientific judgement (HITL, Day 48). And wire one OTEL-to-WORM-style artifact trail that doubles as your reproducibility record, your dual-use audit, and your Annex III evidence.

VIRAL APP OF THE WEEK

FutureHouse Kosmos + the Robin lab-in-the-loop system

The independent counterpart to Claude Science and the AI-for-science tool that went viral this year. FutureHouse (nonprofit, spun its for-profit Edison Scientific out in late 2025) put publicly usable 'AI scientist' agents on the web and an API. Kosmos runs what users estimate is six months of research in a single day: one run reads ~1,500 papers, writes ~42,000 lines of analysis code, and stays coherent across tens of millions of tokens, with roughly 80% of findings judged accurate and 7 validated discoveries across neuroscience, materials science and genetics. Its sibling Robin (Nature, 2026) is the cleanest public example of the day's technique: three language agents -- Crow (fast literature search), Falcon (deep review) and Finch (experimental-data analysis) -- iterate hypothesis -> experiment -> analysis -> refined hypothesis, with human scientists running the physical assays. Applied to dry age-related macular degeneration, Robin proposed boosting RPE-cell phagocytosis, nominated the drug ripasudil, and surfaced ABCA1 as a target -- concept to submitted paper in ~2.5 months. Secondary call-outs: NVIDIA BioNeMo Agent Toolkit (github.com/NVIDIA-BioNeMo/bionemo-agent-toolkit -- 'turn any agent into a life-science expert', now the callable-model layer under both Claude Science and OpenAI) and Ginkgo Bioworks committing all R&D; to autonomous infrastructure by end-2026.

6 months

of research work in a single day
(Kosmos)

42,000

lines of analysis code per run

~2.5 mo

Robin: concept to submitted
drug-discovery paper

~80%

of Kosmos findings judged
accurate

MARKET SIGNAL

With frontier model performance compressed to a near-tie (Day 80), the AI-for-science race is not being won on raw benchmark points -- it is being won on the stack around the model: callable specialist models (BioNeMo), reproducible auditable artifacts, on-prem data boundaries, and reviewer agents that make outputs trustworthy. Claude Science, FutureHouse Kosmos and the BioNeMo toolkit all landed inside two weeks; the moat is the lab-in-the-loop harness plus governance evidence, not the base model. Anthropic, now past OpenAI on revenue (~\$47B run-rate) and courting scientists with a 50-project AI-for-Science fund, is planting the same open-protocol flag in science that MCP planted in tooling.

BREAKING THIS WEEK (JUL 1-6)

Claude Sonnet 5 went GA and, after US Commerce lifted export controls on Jun 30, Fable 5 was restored globally and Mythos 5 re-enabled for select US orgs — capability gating now operating at the export-control layer. OpenAI unveiled a custom 'Jalapeno' inference chip and was reportedly in talks over a ~5% US government stake. Google DeepMind shipped Nano Banana 2 Lite (fastest, cheapest image model) plus Gemini Omni Flash. Blackstone committed \$30B to AI data centres in Japan; Qualcomm acquired Modular for \$4B. And Anthropic — now past OpenAI on revenue (~\$47B run-rate) — opened its Claude Science AI-for-Science fund (up to 50 projects, \$30K credits each + \$2K Modal compute; applications close Jul 15).

PRACTICAL TAKEAWAYS

1. Automate one tight loop, not the scientific method

Pick a step where the experiment is cheap, fast and reversible (literature triage, hypothesis enumeration, an analysis pipeline) and close a hypothesis -> run -> analyse -> refine loop around it. Keep a human on the physical experiment and the 'is it real?' judgement. That is where the Level 2-3 systems actually deliver.

2. Wrap your best models as callable skills

Follow the BioNeMo pattern: expose specialist models (yours or open ones) as MCP skills with crisp descriptions and let a generalist coordinator route to them. Your agent's advantage is knowing WHEN to call Evo 2 or Boltz-2, not trying to be them.

3. Make the artifact trail the product

Ship every result with the code, environment and message history that produced it, and put a reviewer/critic agent in the loop to check citations and numbers. One artifact trail is simultaneously your reproducibility record, your dual-use audit, and your EU AI Act Annex III evidence -- and Aug 2 is T-26.

TOMORROW

Next in the tech pipeline: Self-improving agents 2.0 -- MemRL in production, skill-library evolution, and the Darwin-Godel line -- how an agent gets better without retraining the base model.